

Matthew Jovero

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Seeking an engineering internship full time position starting in 2027.

Education

SAN JOSÉ STATE UNIVERSITY, San José, CA
B.S Aerospace Engineering, Junior

Expected graduation: May 2028
GPA: 3.5/4.0

Work Experience

MATHEMATICS TUTOR | *Self-employment*

Oct 2024 – Jan 2026

- Designed work assignments using LaTeX to guide 7+ clients through solution steps for 3+ engineering mathematics subjects.
- Conducted 1-to-1 instruction to improve student exam performance and build foundational problem-solving strategies.
- Provided online tutoring for international students, adapting to different mathematical notation and terminology differences.

Projects

SJSU 2025 AIAA DESIGN/BUILD/FLY | *Aircraft Design Engineer*

Sept 2024 – May 2025

- Built a MATLAB analysis that supports a 25 lb. aircraft weight, 15 lb. payload, and 70-85 watt-hour battery requirement.
- Designed a 72-inch Clark-Y wing in Inventor with I-beam/tubular spars and lightening holes to correct weight and strength.
- Performed FEA in Ansys to validate internal wing structure when predicting distributed rib loads from 2-8 N.
- Implemented parameter sweeps and generated 3D trade-study graphs evaluating ideal aircraft speeds and power usage.
- Presented performance trade studies and their impact on aircraft design decisions through a formal technical presentation.

AUTONOMOUS VEX ROBOT | *Mechanical Lead*

Mar 2025 – May 2025

- Led a 6-member team to build a robot that autonomously locates and transports IR beacons in 19 seconds (university record).
- Designed a motor-driven mechanical arm using C-channels that maintains strength while satisfying a 2.75 kg weight restriction.
- Programmed RobotC autonomous behaviors such as beacon tracking using sensor feedback and timed state transitions.
- Responsible for mechanical design of the chassis that holds sensors and other control systems to maximize efficiency.

WIND TURBINE DESIGN | *Aerodynamic Design*

Jan 2025 – Mar 2025

- Led the design of a turbine blade in SolidWorks utilizing a NACA 4424 to produce 2 W of power under 10 N/mm stiffness.
- Operated drill presses, saws, and hand tools to prevent wood splitting and ensure part quality for tower fabrication.
- Constructed the final tower using epoxy-bonded engineering wood designed to withstand multidirectional loads of wind.

ATX CASE DESIGN | *Product Design Engineer*

May 2026 – July 2026

- Designed a computer tower for ATX motherboards consisting of 40+ parts in SolidWorks to enhance hardware variety.
- Satisfied user-friendly design factors such as a 120-140 mm fan size, 240 mm radiator, and 25 mm of cable management.

HIGH-POWER ROCKET DESIGN | *Rocketry Designer – Rocket Club*

Oct 2025 – Dec 2025

- Designed a high-power rocket in SolidWorks, reducing cost by 12% when compared to a commercial rocket kit.
- Performed FEA to analyze landing impact loads on the most critical ground-strike fin with PETG polyester material.
- Developed an exploded view diagram and 10 detailed part drawings using GD&T and BOM.
- Fabricated a high-power rocket with a commercial kit using epoxy bonding for an H-class high-power motor.

AHS MACHINE WOODSHOP | *Student Shop Assistant*

Nov 2022 – May 2023

- Involved in training of new students using different machines and equipment safely for their technical projects.
- Maintained condition and organization of the shop and its machines (bandsaws, table saws, drill presses, toolbox, 3D printers).

PANDEMIC SIMULATION MODEL | *Software Development*

Sept 2025 – Dec 2025

- Built a C++/Python pandemic simulation modeling probabilistic disease spread and population health-state transitions.
- Programmed using data structures and modular code enabling updates to simulation using the SIR model.

Skills

Software: SolidWorks, Autodesk Inventor, MATLAB, Python, C++, RobotC, Ansys, FEA, LaTeX, MS Office Suite

Hardware: Mechanical assembly, machine tools, epoxy bonding, system debugging, sensor integration, 3D printing, hand tools